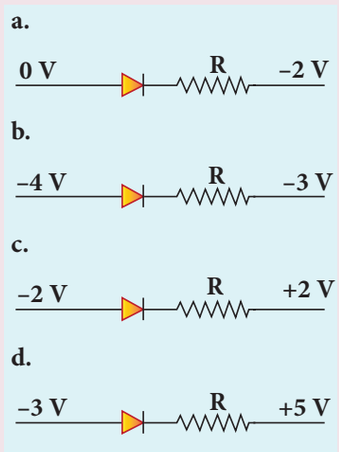
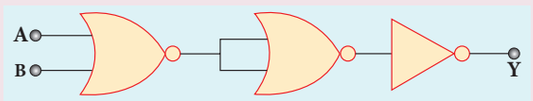


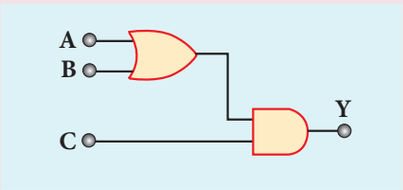
12. The given electrical network is equivalent to (NEET)



a) AND gate b) OR gate

c) NOR gate d) NOT gate

13. The output of the following circuit is 1 when the input ABC is (NEET)



a) 101 b) 100

c) 110 d) 010

14. The variation of frequency of carrier wave with respect to the amplitude of the modulating signal is called

a) Amplitude modulation

b) Frequency modulation

c) Phase modulation

d) Pulse width modulation

15. The frequency range of 3 MHz to 30 MHz is used for

a) Ground wave propagation

b) Space wave propagation

c) Sky wave propagation

d) Satellite communication

Answers

1. a	2. c	3. b	4. c	5. d
6. c	7. a	8. d	9. d	10. a
11. a	12. c	13. a	14. b	15. c

II. Short Answer Questions

1. Define forbidden energy gap.

2. Why is temperature co-efficient of resistance negative for semiconductor?

3. What do you mean by doping?

4. Distinguish between intrinsic and extrinsic semiconductors.

5. A diode is called as a unidirectional device. Explain.

6. What do you mean by leakage current in a diode?

7. Draw the input and output waveforms of a full wave rectifier.

8. Distinguish between avalanche breakdown and Zener breakdown.

9. Give the Barkhausen conditions for sustained oscillations.

10. Explain the current flow in a NPN transistor.

11. What are logic gates?

12. Explain the need for a feedback circuit in a transistor oscillator.

13. Write a short note on diffusion current across p-n junction.

14. What is meant by biasing? Mention its types.

15. Why can't we interchange the emitter and collector even though they are made up of the same type of semiconductor material?

XII_Physics_Vol-2_UNIT-10_Part 2_EM.indd 247 04-03-2022 17:19:26